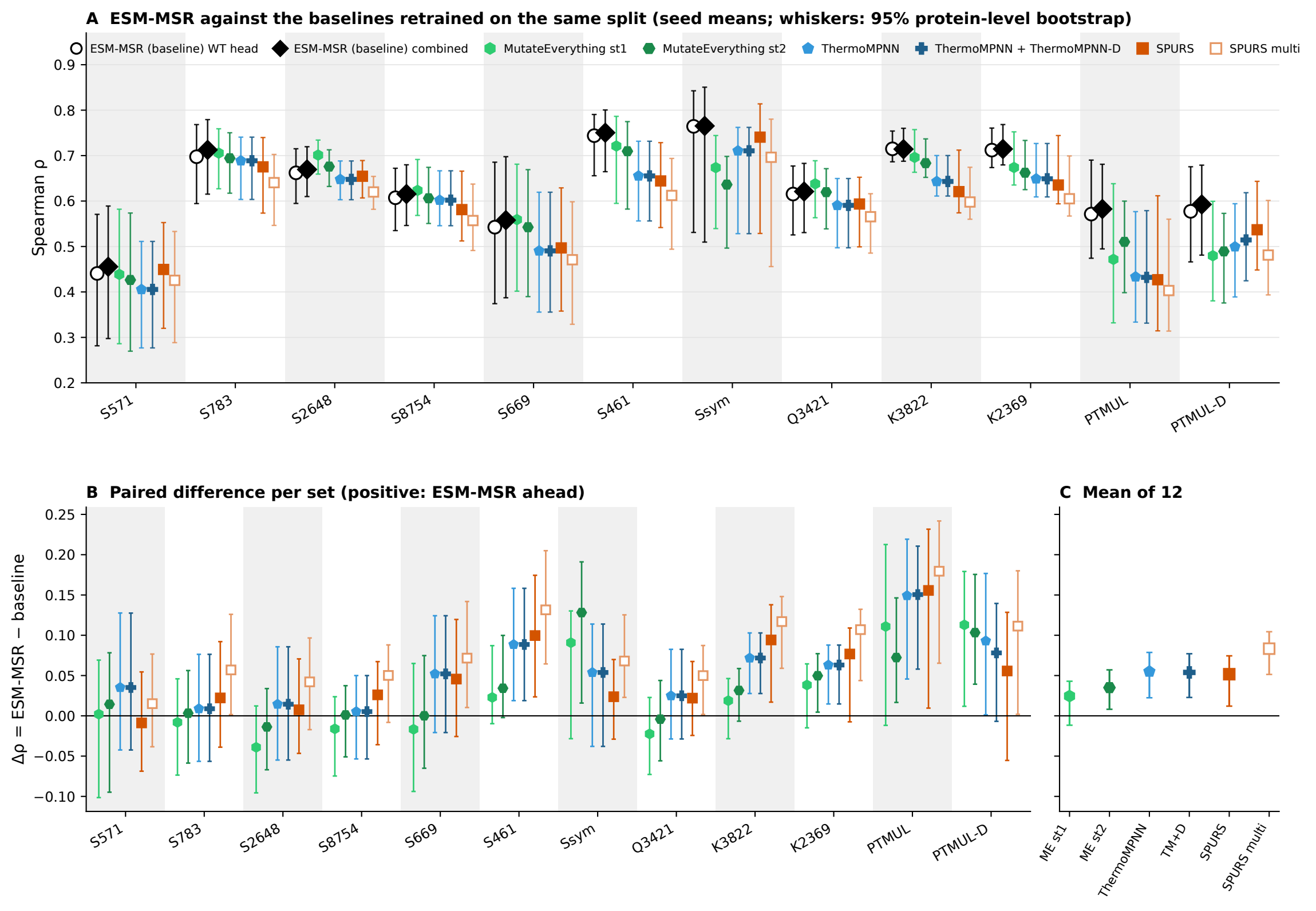
A ESM-MSR against the baselines retrained on the same split (seed means; whiskers: 95% protein-level bootstrap)
ESM-MSR (baseline) WT head
ESM-MSR (baseline) combined
MutateEverything st1
MutateEverything st2
ThermoMPNN
ThermoMPNN + ThermoMPNN-D
SPURS
SPURS multi
Spearman ρ
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
S571
S783
S2648
S8754
S669
S461
Ssym
Q3421
K3822
K2369
PTMUL
PTMUL-D
B Paired difference per set (positive: ESM-MSR ahead)
Δρ = ESM-MSR − baseline
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
S571
S783
S2648
S8754
S669
S461
Ssym
Q3421
K3822
K2369
PTMUL
PTMUL-D
C Mean of 12
ME st1
ME st2
ThermoMPNN
TM+D
SPURS
SPURS multi